Parallelizing the Head-Tracking Pipeline for Webcam-Based Fishtank VR

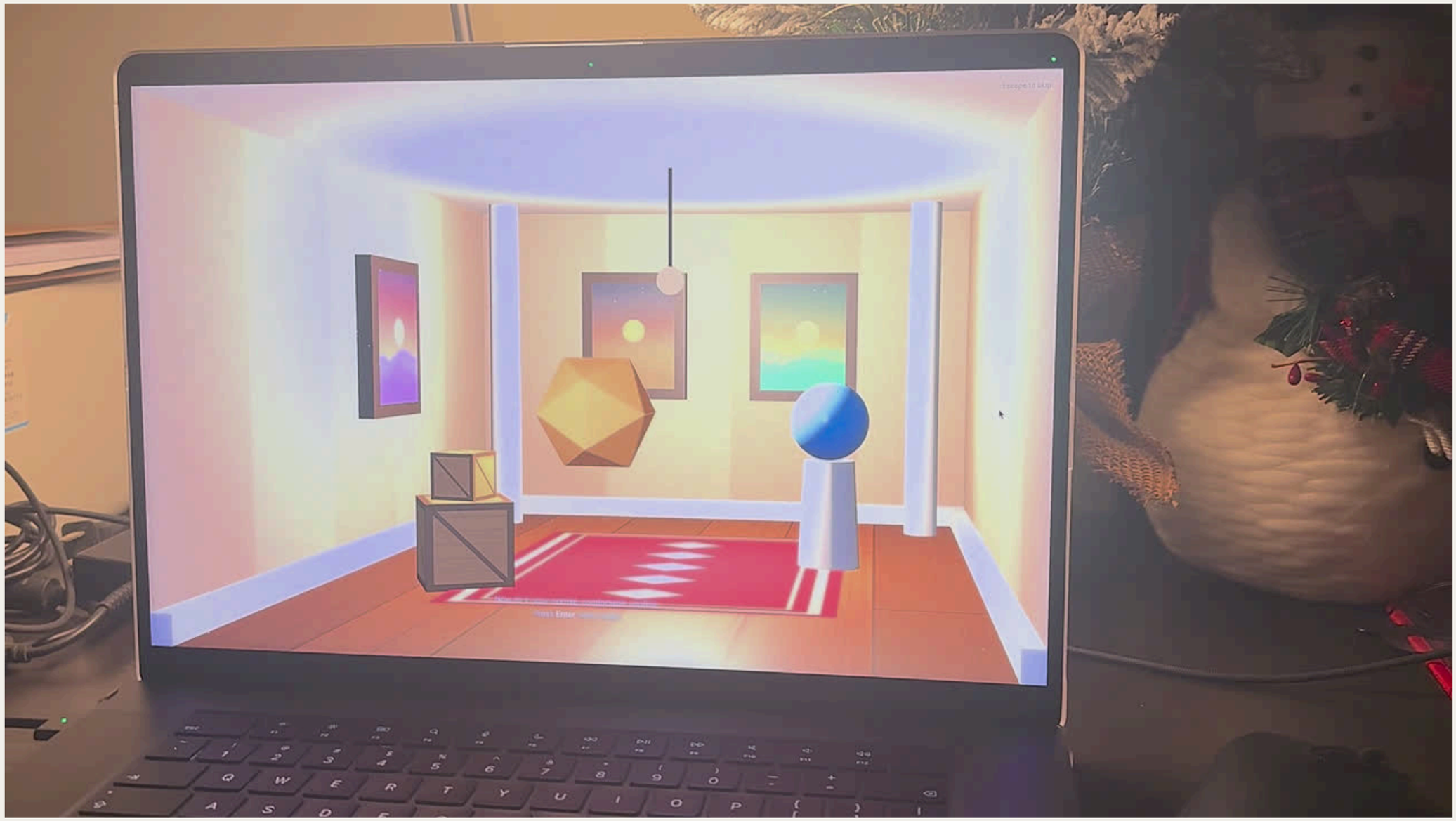
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CSC612M INTEGRATING PROPOSAL

Recap

FISHTANK VR Depth on an ordinary monitor, by tracking your head and re-rendering with an off-axis projection.

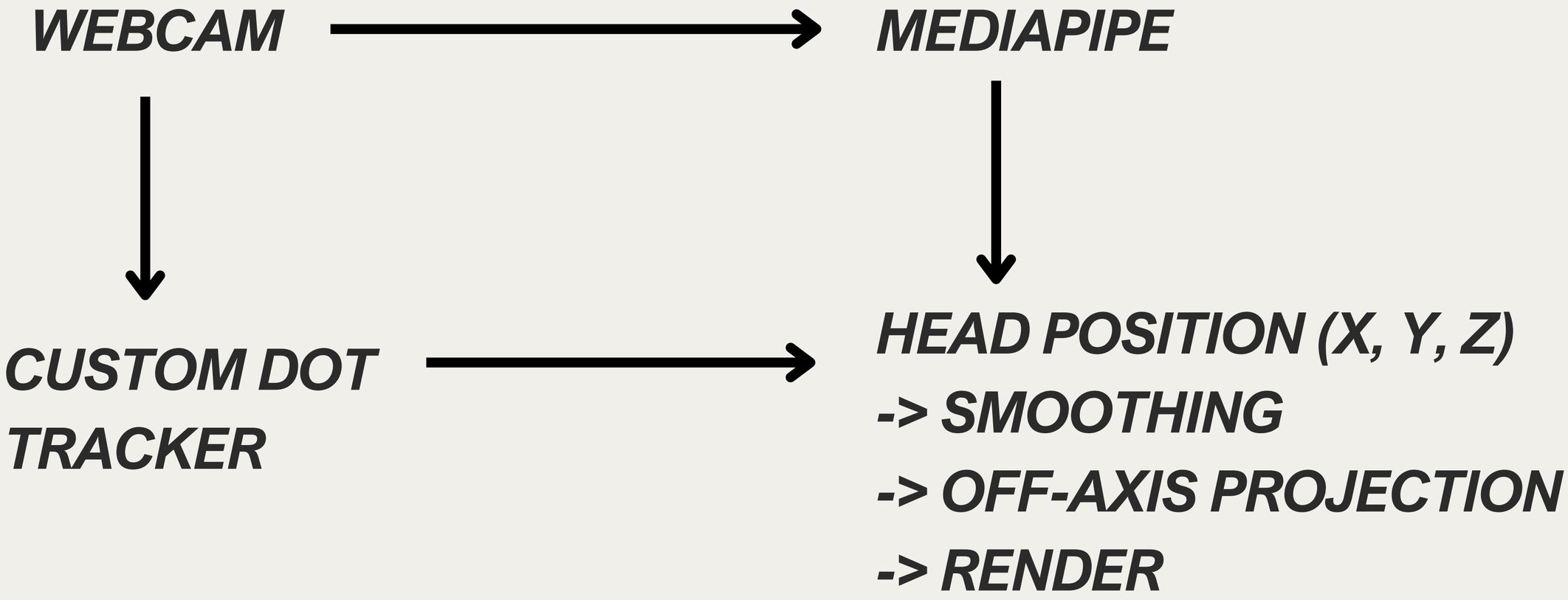
THE CATCH Too much delay between head movement and the updated image breaks the illusion.



PRIMARY GOAL

- ***THE DEMO IS SIMPLY THE TEST BED.***
- ***THE REAL WORK IS MEASURING HOW MUCH SIMD, SIMT, AND PIPELINING CAN SPEED UP THE TRACKING.***

How the system works



Changes

Proposed	Now	Why
IR LED marker	Dotted glasses, regular webcam	No special hardware, anyone can run it
Own OpenGL render scene	Forked open-source engine (Three.js)	Saves time, we focus on the tracker
Single tracker	MediaPipe baseline + our dot tracker	Working demo now, plus a comparison point

Milestone Status

Proposal item	Status
Webcam capture	**Finished**
Working display / emulator	**Finished**
Scalar C dot tracker	**Started**
Timing code	**Started**
Datasets + correctness checker	Not Started
AVX2 kernels (S1 to S4)	Not Started
Unfused CUDA kernels	Not started
Fused versions, pinned memory, streams	Not started
RISC-V kernel + hazard analysis	Not Started

Thank You